



Identification and Quantification of Carotenoids and Tocochromanols in Sorghum Grain by High-Performance Liquid Chromatography

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Abstract

Carotenoids and tocochromanols are lipid-soluble secondary plant metabolites that are essential for the normal functioning of plants and, in some cases function as a source of vitamin A and E in humans. Enhancement of the provitamin A carotenoid and tocochromanols levels in sorghum and other cereal grains through traditional breeding and transgenic methods has increased in recent years with interest in biofortification of grains to combat micronutrient deficiencies in developing countries. With this increase in research, reliable methodology for the extraction, identification, and quantification of individual carotenoids and tocochromanols species from sorghum and other cereal grains is essential. Here, we describe a basic method for extraction of carotenoid and tocochromanols adapted to sorghum grain and chromatographic condition for separation, identification, and quantification of individual carotenoid using High-Performance Liquid Chromatography (HPLC) system.

Key words Carotenoids, Vitamin A, Vitamin E, Lutein, Zeaxanthin, β -carotene, Tocopherols, Tocotrienols

1 Introduction

Carotenoids are natural pigments, responsible for the yellow, orange, and red coloration of many commonly consumed fruits, vegetables, and grains. The rich color of these molecules is a result of their tetraterpene structure with an extensive system of conjugated double bonds [1]. Carotenoids are generally classified into two major groups: carotenes, which are strictly hydrocarbons, that are either cyclized such as β -carotene, or linear, such as lycopene; xanthophylls, or oxycarotenoids, which contain oxygen that include lutein, zeaxanthin, and b-cryptoxanthin [2]. Dietary carotenoids are those carotenoids in the food chain, and their number has been estimated to be approximately 40–50; however, only 24 of them have been reported in human plasma and tissues [3, 4]. Consumption of these phytochemicals has been associated with specific

health benefits such as reduced risk of cardiovascular diseases, certain type of cancer, Alzheimer's disease, and eye-related maculopathy [5–8]. Lutein, zeaxanthin, and the lutein metabolite meso-zeaxanthin accumulate in the macula of the human eye and are vital for ocular health [9]. Many health benefits are attributed to carotenoids and their role in modulating inflammatory response [10], and oxidative stress [11]; ability to regulate gene transcription, gap junction communication, phase II enzyme-inducing activity, and ability to enhance immune function [12]. While promising, provitamin A activity remains the best-known role of carotenoids in human health [13]. Nutritionally, α -, β -, γ -carotene as well as α - and β -cryptoxanthin are considered provitamin A carotenoids (pVACs). These carotenoids have at least one unsubstituted β -ionone ring, which when cleaved in the intestine or liver leads to the production of retinol (vitamin A) [14]. With dietary retinol forms being derived primarily from animal or supplemental sources, pVACs remain the main plant-derived dietary form and therefore critical to delivery of vitamin A to many at-risk populations globally. As such, efforts to develop and implement new sorghum genotypes with enhanced pVAC levels are under way to increase the availability of provitamin A and other beneficial carotenoids in the human diet.

Similar to carotenoids, tocochromanols are a group of amphipathic, lipid-soluble organic molecules composed of a polar moiety derived from tyrosine and a hydrophobic polyprenyl side chain originating from the isoprenoid pathway [15]. Tocochromanols are grouped into two classes, tocopherols characterized by a phytyl-derived side chain whereas tocotrienols are those with a geranylgeranyl-derived side chain. The four different forms of tocopherols and tocotrienols (α -, β -, γ and δ -) differ by the degree of methylation of the polar moiety [15]. As such, four tocopherols and four tocotrienols constitute natural vitamin E forms found in food. However, the humans preferentially use α -tocopherol as vitamin E and only this form can reverse deficiency symptoms and, by extension, meet human vitamin E requirements [16]. α -tocopherol's major vitamin functions are to maintain the integrity of long-chain polyunsaturated fatty acids in the membrane cells and thus maintain their bioactivity acting as a potent peroxyl radical scavenger [17]. Vitamin E is an essential nutrient in the human diet that can only be produced by photosynthetic organisms. Diets that are low in α -tocopherol may be at risk for increased oxidative damage because they have insufficient protection from oxidative stress [16]. While α -tocopherol is the most relevant vitamin E compound, it is not the main form consumed since the composition of most major crops is dominated by γ -tocopherol [18]. Therefore, several strategies have been undertaken to improve vitamin E content in crops. In addition, enhancing the levels of vitamin E might enhance the accumulation and stability of

provitamin A carotenoids in biofortified sorghum [19]. Hence, a method for simultaneous analysis of vitamin E and carotenoids isomers is essential in plant breeding programs; therefore, here it is described an optimized method for the identification and quantification of carotenoids and tocochromanols in sorghum grain by HPLC.

2 Materials

Prepare all solutions using ultrapure water (prepared by purifying deionized water, to attain a sensitivity of 18 MΩ-cm at 25 °C). All sample preparations and extractions must be performed under low light conditions or yellow/red light to minimize potential for photo-isomerization reaction.

1. Extraction solvents: Acetone and Methyl tert-butyl ether. It is recommended their use in American Chemical Society (ACS)-grade and fresh or the addition of butylated hydroxytoluene (BHT) (0.1%) as a stabilizer agent.
2. Mobile phase solvents: Methanol and Ethyl Acetate. It is recommended their use in HPLC-grade.
3. Ammonium acetate: 1.0 M Ammonium acetate must be prepared using double distilled water and adjusted to pH 4.6 with glacial acetic acid.
4. Saturated saline solution. Solubility of NaCl in water is 36 g per 100 mL at 20 °C. Prepare the saturate saline solution by slowly adding sodium chloride (ACS-grade) to a double distilled water and stir the mixture thoroughly.

3 Methods

3.1 Sample Preparation

1. Collection of sorghum grain from field plots should be conducted based on appropriate sampling and statistical plan. Sufficient samples should be collected to ensure reflection of natural variation and replication of analysis.
2. In the laboratory, take collected grain head and dehull them. Pile the grains (approximately 500 g–1 kg) evenly on a clean surface, flatten the pile and spread into a circle.
3. Make a cross, dividing the circle into four roughly equal parts. Discard two diametrically opposite quarters and remix the remaining two quarters.
4. Repeat the quartering procedure until the amount is reduced to approximately 125–250 g.
5. Grind the grains to <0.5 mm and keep the sorghum meal in a tightly closed container stored frozen at –20 °C.

3.2 Extraction

1. Keep ground sorghum meal on ice while weighing and covered (*see Note 1*).
2. Weigh ~600 mg of ground sorghum flour in a 15 mL Falcon tube, polypropylene. Keep weighed samples on regular ice in covered styrofoam cooler. Keep covered to reduce light.
3. Spike samples at this point with 80 μ L of internal standards β -apo-8-carotenal and echinenone (resolubilized in 1 mL Petroleum ether). Run 2 spikes and one control samples for each extraction set. Also, spikes should be in a range of 1X–3X in concentration to the highest carotenoid present. Inject 10 μ L.
4. Add 1.0 mL of saturated saline solution. Vortex to mix. Place on ice in covered cooler for 10 min.
5. Add 5 mL of chilled acetone (with 0.1% butylated hydroxytoluene). Vortex to mix. Agitate for 5 min on rotary mixer. Wrap sample rack in foil to reduce light. Incubate for 5 min on ice in covered ice cooler (*see Note 2*).
6. Centrifuge samples at 3000 RPM ($1734 \times g$) for 3 min cooled to 4 °C.
7. Remove samples from centrifuge and transfer acetone into a second 15 mL test tube.
8. Evaporate acetone fraction under nitrogen gas.
9. Add a 5 mL aliquot of chilled acetone to grain sample. Vortex to mix pellet. Agitate for 5 min on rotary mixer (Vortex-Genie 2, 3000 RPM). Incubate for 5 min on ice. Centrifuge at 3000 RPM ($1734 \times g$) for 3 min at 4 °C. Add this second acetone fraction to the initial acetone fraction under nitrogen gas.
10. Add 2 mL of methyl tert-butyl ether (with 0.1% butylated hydroxytoluene) to ground sample. Vortex to mix pellet. Agitate for 5 min on rotary mixer. Incubate for 5 min on ice in a covered cooler. Vortex for 5 min on a rotary mixer (Vortex-Genie 2, 3000 RPM).
11. Centrifuge samples at 3000 RPM ($1734 \times g$) for 3 min at 4 °C.
12. Mix the methyl tert-butyl ether (lipophilic) fraction with the acetone (aqueous) fraction that was previously evaporated under nitrogen gas (*see Note 3*).
13. Add a 1 mL of methyl tert-butyl ether and gently add 1 mL saturated saline solution. Gently mix the mixture and wait until a very well-defined phase separation occurs in the mixture.
14. Collect the upper phase with a Pasteur glass pipette avoiding the lower phase and transfer it to a new 15 mL test tube. Repeat this **step 3** more times or until no visible color appears

in the upper phase of the mixture. Place the new 15 mL test tube with all the lipophilic fractions under nitrogen gas to completely evaporate the extracts.

15. Redissolve the dried extract in 2 mL of ethyl acetate:methanol (1:1) for immediate analysis. Vortex briefly to mix sample and syringe filter through 0.45 mm filters prior to injection.

3.3 Chromatographic Separation

Carotenoids and tocochromanols analysis is recommended to be performed in a HPLC couple with a multichannel UV-Visible detector or Photodiode Array Detector (PDA or DAD detector). A PDA is preferred as it allows for the acquisition of the UV-Visible spectra in-line, assisting in the identification of main carotenoids and tocopherols and providing a means of verifying peak purity (absence of interfering compounds).

Column type:	YMC C30 3 μ m 2.0 mm $\times$ 150 mm column, with a YMC carotenoid guard column (2.0 $\times$ 23 mm)
Column temperature:	40 $^{\circ}$ C
Mobile phase:	Binary mobile phase system. Phase A: Methanol:1 M ammonium acetate (98:2 v/v). Phase B: Ethyl acetate
Flow:	0.37 mL/min
Detection:	UV-Visible or PDA, 295 nm, 348 nm, and 450 nm.
Injection:	10 μ L
Internal standard:	<i>trans</i> - β -apo-8-carotenal (CAS Number: 1107-26-2), echinenone (CAS Number: 80348-65-8), and DL α -tocopherol acetate (CAS Number: 7695-91-2)
Elution gradient:	0 min 80:15 v/v, phase A:phase B; 6 min 20:80 v/v, phase A:phase B; 8 min 0:100 v/v, phase A:phase B; 12 min 0:100 v/v, phase A:phase B, 14 min 80:15 v/v, phase A:phase B
Post-injection time:	3 min to equilibrate between runs

3.4 Identification

While more advanced mass spectroscopy methods have been recently applied to the characterization of carotenoids and tocochromanols in foods [20, 21], tentative identification is most commonly carried out by co-chromatography with authentic standards and comparison of ultraviolet and visible absorption spectrum. The commercial availability of many all-*trans* carotenoid standards has made this possible. Maximum absorption peak (λ_{max}) and the shape (spectral fine structure) of the spectrum can be used as a tool for carotenoids identification (Fig. 1) (*see Note 4*). Characteristics λ_{max} values of major carotenoid species in sorghum are presented in Table 1. A typical carotenoids separation from transgenic sorghum meal is shown in Fig. 2.

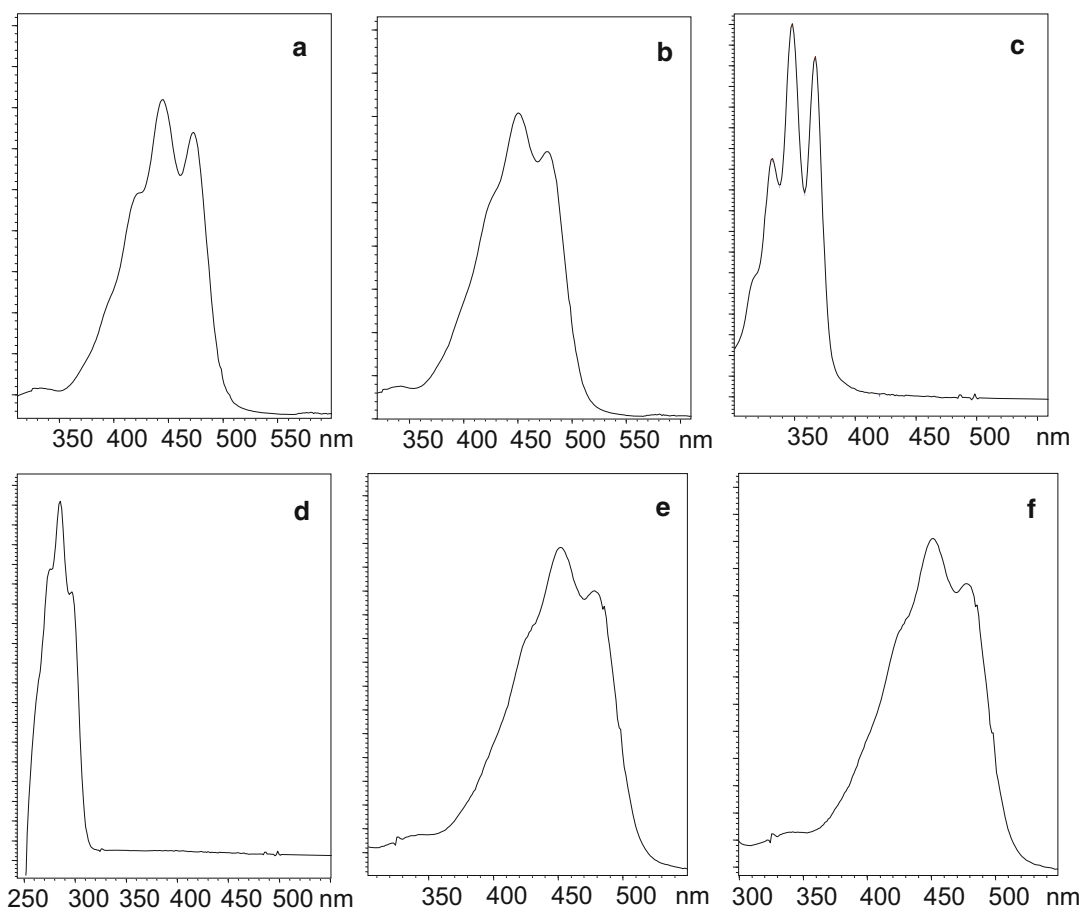


Fig. 1 UV-Visible absorption spectra collected in-line during LC analysis of carotenoid species in sorghum meal. Peak identification: lutein (a), zeaxanthin (b), phytofluene (c), phytoene (d), all-trans- β -carotene (e), β -cryptoxanthin (f)

3.5 Quantification

Carotenoids and tocochromanols in solution obey the Beer-Lamber law; therefore, their absorbance is directly proportional to the concentration. Thus, carotenoids and tocochromanols are typically quantified spectrometrically for generation of stock solutions suitable for calibration curves and compatible with typically UV-Visible detection. In this regard, HPLC quantification is most commonly carried out by generation of an external calibration curve with the respective standard. Concentrations of standard solutions are determined by visible absorption spectrometry. Table 2 shows the absorption coefficients of the major carotenoids and tocochromanols present in sorghum meal. Following preparation of standards, a calibration curve is constructed by injecting a range of standard concentrations (see below) on the HPLC from which the peak areas are plotted against the concentration. Using linear regression, a calibration curve can be established from which carotenoid concentrations can be calculated from peak areas obtained from sample analysis.

Table 1
Ultraviolet and visible absorption data for major carotenoid species sorghum meal

Carotenoid	Solvent	λ_{max} , nm ^a			% III/II ^b
α -Carotene	Acetone	424	448	476	55
	Chloroform	433	457	484	
	Ethanol	423	444	473	
	Hexane, petroleum ether	422	445	473	55
β -Carotene	Acetone	(429)	452	478	15
	Chloroform	(435)	461	485	
	Ethanol	(425)	450	478	25
	Hexane, petroleum ether	(425)	450	477	25
α -Cryptoxanthin	Chloroform	435	459	487	
	Ethanol	423	446	473	60
	Hexane	421	445	475	60
β -Cryptoxanthin	Chloroform	(435)	459	485	
	Ethanol	(428)	450	478	27
	Petroleum ether	(425)	449	476	25
Lutein	Chloroform	435	458	485	
	Ethanol	422	445	474	60
	Petroleum ether	421	445	474	60
Phytoene	Hexane, petroleum ether	(276)	286	(297)	
Phytofluene	Hexane, petroleum ether	331	348	367	
Zeaxanthin	Acetone	(430)	452	479	
	Chloroform	(433)	462	493	
	Ethanol	(428)	450	478	26
	Petroleum ether	(424)	449	476	25

Adapted from: Rodriguez-Amaya et al. [22]

^aParenthesis indicate a shoulder

^bRatio of the height of the longest-wavelength absorption peak, designated III, and that of the middle absorption peak, designated II, taking the minimum between the two peaks as baseline, multiplied by 100

3.5.1 External Calibration Procedure

1. Turn on UV/vis spectrophotometer lamps.
2. Dissolve a small quantity (~0.5–1 mg) of authentic standard into a culture tube.
3. Add 10–15 mL of the appropriate solvent according (Table 2) to and sonicate for 5 min.
4. Filter with 0.45 μm syringe filter and label tube A.

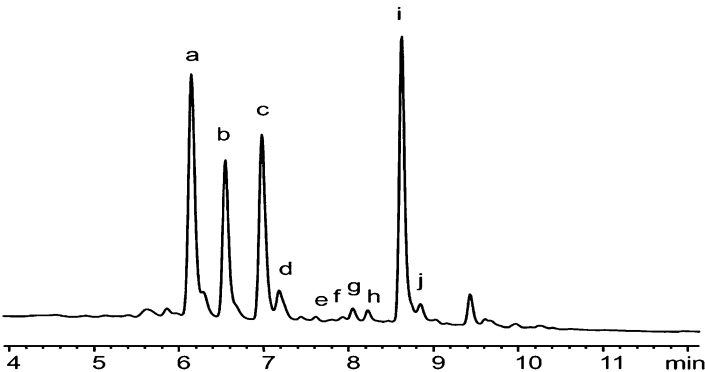


Fig. 2 Chromatogram of major carotenoid species in transgenic sorghum meal (Hemi203-A) analyzed by HPLC with diode-array detection at 450 nm. Peak identification: lutein (**a**), zeaxanthin (**b**), β -apo-8'-carotenal internal standard (**c**), α -cryptoxanthin (**d**), β -cryptoxanthin (**e**), 15-cis- β -carotene (**f**), 13-cis- β -carotene (**g**), α -carotene (**h**), all-trans- β -carotene (**i**), 9-cis- β -carotene (**j**). Reprinted with permission from Lipkie et al. [23] (Copyright 2017, American Chemical Society)

Table 2
UV absorptivities of carotenoids and tocochromanols

Carotenoid	Wavelength (nm)	$A^{1\%}_{1\text{cm}}$	Solvent
β -carotene	450	2592	Petroleum Ether
Lycopene	470	3450	Petroleum Ether
α -carotene	445	2710	Hexanes
β -cryptoxanthin	450	2460	Hexanes
α -cryptoxanthin	445	2636	Hexanes
Lutein	445	2550	Ethanol
Zeaxanthin	449	2348	Petroleum Ether
Phytoene	286	1250	Petroleum Ether
Phytofluene	348	1350	Petroleum Ether
Tocochromanols	Wavelength (nm)	$A^{1\%}_{1\text{cm}}$	Solvent
α -Tocopherol	292	75.8	Ethanol
β -Tocopherol	296	89.4	Ethanol
γ -Tocopherol	298	91.4	Ethanol
δ -Tocopherol	298	87.3	Ethanol
α -Tocotrienol	290	91	Ethanol
β -Tocotrienol	295	87.5	Ethanol
γ -Tocotrienol	298	103	Ethanol
δ -Tocotrienol	298	83	Ethanol

Source: Britton G [24] and Podda et al. [25]

Table 3
Schematic dilution procedure for curve calibration preparation

Sample	Dilution	Sample added	Solvent
A	1		
B	1/2	5 mL A	5 mL
C	1/10	2 mL B	8 mL
D	1/50	2 mL C	8 mL
E	1/100	1 mL C volumetrically	In 10 mL volumetric flasks
F	1/500	1 mL D volumetrically	In 10 mL volumetric flasks
G	1/1000	1 mL E volumetrically	In 10 mL volumetric flasks

5. Read absorbance, adjust to ~1 with appropriate solvent.
6. Dilute following the scheme (Table 3) for A–D. Vortex and cap.
7. Read absorbance of three different 1 mL aliquots each with new cuvette.
8. Prepare E–G (Table 3) volumetrically using volumetric flask and volumetric pipets.
9. If solvent is hexane transfer 2 mL volumetrically of each sample into a new culture tube.
 - (a) Dry with nitrogen.
 - (b) Resolubilized volumetrically 1 mL EtOAc and 1 mL MeOH.
10. If methanol, transfer directly to HPC vial.
11. Inject 10 μ L.

4 Notes

1. Extraction should be carried out under a yellow light to avoid photoisomerization reactions. Solvents used during the extraction procedure must be chilled and stabilized with butylated hydroxytoluene (BHT) (0.1%).
2. Since water was added at the beginning of the extraction process, and water is soluble in the acetone fraction, therefore the resulting fraction will be a mixture of acetone and water.
3. Recommendation. Add the methyl tert-butyl ether fraction to the acetone fraction placed previously under nitrogen after you notice that carotenoids begin to float on the surface of the solution or foaming formation on the surface of the solution. This phenomenon is an indication that most of the acetone in

the mixture has been evaporated and the composition of the remaining solution is mainly water.

4. Identification of carotenoids based solely on the retention times/co-chromatographic with standards or the absorption spectra may lead to erroneous conclusions [22]. However, carotenoids with well-known structure can be conclusively identified by the combined and judicious use of chromatographic behavior, UV-visible absorption spectra, and for the xanthophyll's, specific group chemical reactions [26].

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